Lecture 3

Homotopy and the fundamental group

Objectives. *We compare paths by continuous deformations that keep their endpoints fixed. A path returning to its starting point is a loop. The classes of loops at one fixed point form the fundamental group. For the circle this group is the integers: a continuously chosen angle records how many turns a loop makes. We prove this by lifting paths and homotopies, then use the integer to prove the fundamental theorem of algebra.*

By the end of this lecture you will be able to:

- Construct the fundamental group, induced homomorphisms and base-point isomorphisms.
- Prove that lifting computes the fundamental group of the circle as $\mathbb{Z}$.
- Compute the group of the punctured plane and prove the disc has no retraction onto its circle.
- Prove that a nonconstant complex polynomial has a root.

3.1 Homotopy and fixed endpoints

Write $I = [0, 1]$. Recall that a path in X is a continuous map $\gamma : I \to X$ (§ 2.3). A *loop* has equal endpoints. Their common value is its *base point*.

Definition 3.1 (Homotopy). A *homotopy* from $f : X \to Y$ to $g : X \to Y$ is a continuous map $h : X \times I \to Y$ with $h(x, 0) = f(x)$ and $h(x, 1) = g(x)$ for every $x \in X$. We write $f \simeq g$. For paths from x_0 to x_1, a homotopy *relative to the endpoints* also satisfies

$$h(0, s) = x_0, \qquad h(1, s) = x_1 \qquad (s \in I).$$

Write $h_s(x) = h(x, s)$ for a fixed-parameter map in the homotopy. Continuity is required jointly in the path parameter t and deformation parameter s. For example, the paths $\gamma(t) = (t, 0)$ and $\delta(t) = (t, t(1 - t))$ in $\mathbb{R}^2$ are homotopic relative to their endpoints by

$$h(t, s) = (t, s\, t(1 - t)).$$

This map is continuous because its coordinates are polynomials. At $t = 0$ it is $(0, 0)$ and at $t = 1$ it is $(1, 0)$, independently of s. The figure shows the intermediate paths.

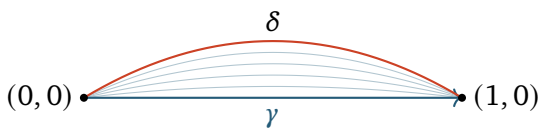


Figure 3.1: The homotopy keeps both endpoints fixed. Increasing s moves the straight path γ to the parabola δ.

We will glue formulas on finitely many closed pieces. If continuous maps on a finite closed cover agree on overlaps, their union is continuous. Indeed, the preimage of a closed set is a finite union of sets closed in those pieces, hence closed in the whole domain. This is the pasting argument of § 2.3, now also applied to squares.

Homotopy relative to endpoints is an equivalence relation. The constant family proves reflexivity, and $h(t, 1-s)$ reverses a homotopy. To join two homotopies, run the first on $0 \le s \le 1/2$ and the second on $1/2 \le s \le 1$, rescaling each parameter interval to I. They agree at the common edge, so the pasting argument proves transitivity.

3.2 The fundamental group

For paths γ from x to y and δ from y to z, define

$$(\gamma \cdot \delta)(t) = \begin{cases} \gamma(2t), & 0 \le t \le \frac{1}{2}, \\ \delta(2t-1), & \frac{1}{2} \le t \le 1. \end{cases}$$

The convention is γ **first, then** δ, as in Hatcher, §1.1. Define the reversed path by $\bar{\gamma}(t) = \gamma(1-t)$, and the constant path by $e_x(t) = x$. All these paths are continuous, by pasting at the matching endpoints. Write $[\gamma]$ for the endpoint-fixed homotopy class of γ.

Theorem 3.2 (Fundamental group). *The endpoint-fixed homotopy classes of loops at $x_0 \in X$ form a group, written $\pi_1(X, x_0)$, with*

$$[\gamma][\delta] = [\gamma \cdot \delta], \qquad 1 = [e_{x_0}], \qquad [\gamma]^{-1} = [\bar{\gamma}].$$

Proof. Products respect classes. Concatenate two endpoint-fixed homotopies in the t direction; their formulas agree along $t = 1/2$. Reversing the t parameter similarly respects classes.

A change of timing does not change a path class. For any continuous $\rho\colon I \to I$ with $\rho(0) = 0$ and $\rho(1) = 1$, the formula $h(t,s) = \gamma((1-s)\rho(t) + st)$ joins $\gamma \circ \rho$ to γ while fixing the endpoints. For associativity, the following timing changes $((\gamma \cdot \delta) \cdot \eta) \circ \rho$ into $\gamma \cdot (\delta \cdot \eta)$:

$$\rho(t) = \begin{cases} t/2, & 0 \le t \le \frac{1}{2}, \\ t - \frac{1}{4}, & \frac{1}{2} \le t \le \frac{3}{4}, \\ 2t - 1, & \frac{3}{4} \le t \le 1. \end{cases}$$

The formulas agree at the break times and fix 0 and 1. On the first interval the parameter in γ becomes $4\rho(t) = 2t$. On the middle interval it becomes $4\rho(t) - 1 = 4t - 2$ in δ. On the last it becomes $2\rho(t) - 1 = 4t - 3$ in η. These are precisely the timings of $\gamma \cdot (\delta \cdot \eta)$. The preceding homotopy therefore proves associativity of classes.

The paths $e_x \cdot \gamma$ and $\gamma \cdot e_y$ are γ with timings $\max(2t-1, 0)$ and $\min(2t, 1)$. Thus their classes equal $[\gamma]$. Finally, the homotopy

$$h(t,s) = \gamma\big((1-s)\min(2t, 2-2t)\big)$$

contracts $\gamma \cdot \bar{\gamma}$ to e_x, fixing both endpoints. Applying it to $\bar{\gamma}$ gives the other inverse identity. These homotopies also apply to joined paths with different endpoints. ∎

A *homomorphism* preserves group products. A bijective homomorphism is an *isomorphism*. A continuous map $f\colon X \to Y$ induces

$$f_*\colon \pi_1(X, x_0) \longrightarrow \pi_1(Y, f(x_0)), \qquad [\gamma] \longmapsto [f \circ \gamma].$$

Proposition 3.3 (Induced homomorphisms). *The induced map f_* is a homomorphism. Induced maps preserve composition and identity maps.*

Proof. Composing a path homotopy with f proves well-definedness. Composition also respects the two path pieces, so f_* preserves products. For continuous $g\colon Y \to Z$, this gives $(g \circ f)_* = g_* \circ f_*$. Identity maps give $(\mathrm{id}_X)_* = \mathrm{id}$. ∎

The base point requires care. For a path a from x_0 to x_1, the base-point change map is

$$c_a\colon \pi_1(X, x_0) \to \pi_1(X, x_1), \qquad c_a([\gamma]) = [\bar{a} \cdot \gamma \cdot a].$$

The figure gives the traversal order.

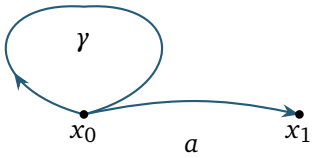


Figure 3.2: A loop at x_1 follows $\bar{a}$ to x_0, traverses γ, then follows a back.

Proposition 3.4 (Change of base point). *For every path a from x_0 to x_1, the map c_a is an isomorphism.*

Proof. Concatenating with fixed paths respects homotopy, so c_a is well-defined. Cancellation of $a \cdot \bar{a}$ proves that c_a preserves products. The inverse is $c_{\bar{a}}$, by the same cancellation. Thus it is an isomorphism. ∎

Elements u and v of a group are *conjugate* if $v = k^{-1}uk$ for some element k of that group. When $x_0 = x_1$, the map c_a is conjugation by $[a]$. If $[a][\gamma] \neq [\gamma][a]$, then $c_a([\gamma]) \neq [\gamma]$, whereas the constant choice of a gives the identity map. Thus the isomorphism can depend on the chosen path.

In a *free homotopy of loops*, each h_s is a loop, but its base point may move. The defining endpoint condition is

$$h(0, s) = h(1, s) \qquad (s \in I).$$

Let $\gamma_0 = h_0$ be based at x_0, and put $\gamma_1 = h_1$. The base point follows the path $a(s) = h(0, s)$. Put $a_s(t) = a(st)$. The loops $a_s \cdot h_s \cdot \bar{a}_s$ keep base point x_0. At $s = 0$, the inserted constant paths do not change the class, by the unit laws. At $s = 1$ this gives $[\gamma_0] = [a \cdot \gamma_1 \cdot \bar{a}]$. If γ_1 is also based at x_0, then a is a loop at x_0. The equality therefore makes $[\gamma_0]$ and $[\gamma_1]$ conjugate.

Conversely, let a be a loop at x_0. The family $\bar{a}_s \cdot \gamma \cdot a_s$ consists of loops with base point $a(s)$. Its end loops represent $[\gamma]$ and $[\bar{a} \cdot \gamma \cdot a]$. Both families are continuous by pasting their three path pieces. Free homotopy is an equivalence relation by the same pasting argument as before. Based homotopies are free homotopies, so conjugate classes are freely homotopic. Thus two loops based at x_0 are freely homotopic exactly when their classes are conjugate. In a path-connected space, move any base point to x_0 by a path. Free homotopy classes then correspond to conjugacy classes in $\pi_1(X, x_0)$, rather than to individual group elements.

3.3 Lifting phases on the circle

Identify S^1 with the unit complex numbers, and set

$$p\colon \mathbb{R} \to S^1, \qquad p(u) = e^{2\pi i u}.$$

We have $p(u) = p(v)$ exactly when $u - v \in \mathbb{Z}$. For a continuous map $f\colon Y \to S^1$, a *lift* through p is a continuous map $\tilde{f}\colon Y \to \mathbb{R}$ with $p \circ \tilde{f} = f$. For this projection, a lift is also called a *phase*.

For $r \in \mathbb{R}$ put

$$J_r = (r - 1/4, r + 1/4), \qquad U_r = \{z \in S^1 : \operatorname{Re}(p(-r)z) > 0\}.$$

These open semicircles cover the circle. The restriction $p\colon J_r \to U_r$ has the continuous inverse branch

$$b_0(z) = \frac{1}{2\pi} \arctan \frac{\operatorname{Im} z}{\operatorname{Re} z}, \qquad b_r(z) = r + b_0(p(-r)z).$$

For $z = e^{i\theta} \in U_0$ with $|\theta| < \pi/2$, the quotient $\operatorname{Im} z / \operatorname{Re} z$ is $\tan\theta$. The principal arctangent returns θ, so $b_0(z) = \theta/(2\pi) \in J_0$ and $p(b_0(z)) = z$. Conversely, for $u \in J_0$ the angle $2\pi u$ is in this range, so $b_0(p(u)) = u$. Rotating gives the inverse on U_r:

$$p(b_r(z)) = p(r)\, p(-r) z = z \qquad (z \in U_r).$$

The figure shows repeated real intervals for one semicircle.

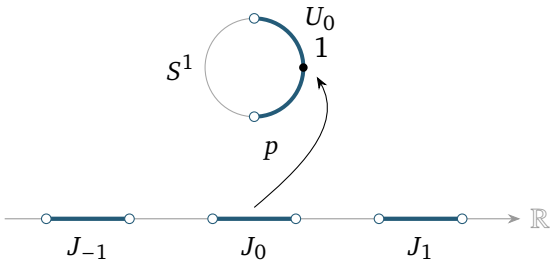


Figure 3.3: The intervals J_{-1}, J_0 and J_1 map to the same open semicircle U_0. Its boundary points are excluded.

We first justify the finite subdivisions used below. The *diameter* of a bounded nonempty set is the supremum of its pairwise distances.

> **Lemma 3.5** (Lebesgue number). *An open cover of a nonempty compact metric space K has a number $\delta > 0$ such that each nonempty subset of diameter less than δ lies in one cover member.*

Proof. For each $x \in K$ choose $\varepsilon_x > 0$ with $B_{2\varepsilon_x}(x)$ inside a cover member. Finitely many balls $B_{\varepsilon_x}(x)$ cover K; let δ be the minimum of their radii. For a set A of diameter less than δ, choose $a \in A$ and one selected ball $B_{\varepsilon_x}(x)$ containing a. For every $z \in A$,

$$d(z, x) \le d(z, a) + d(a, x) < \delta + \varepsilon_x \le 2\varepsilon_x.$$

Thus A lies in the associated cover member. ∎

> **Lemma 3.6** (Path and homotopy lifting for the circle). *A path $\gamma\colon I \to S^1$ has a unique lift after specifying $\tilde{\gamma}(0) \in p^{-1}(\{\gamma(0)\})$. A homotopy $h\colon I^2 \to S^1$ has a unique lift after a lift of its bottom edge $h(t, 0)$ has been specified.*

Proof. The interval and square are compact by Heine–Borel. Apply Lemma 3.5 to the cover of I by $\gamma^{-1}(U_r)$. For its Lebesgue number δ, choose N with $1/N < \delta$ and put $t_i = i/N$ for $i = 0, \dots, N$. Each subinterval maps into some U_r. On $[t_i, t_{i+1}]$, set $\tilde{\gamma} = b_r \circ \gamma + k$, with

$$k = \tilde{\gamma}(t_i) - b_r(\gamma(t_i)) \in \mathbb{Z}.$$

This matches the value already assigned at t_i. The pieces agree at their endpoints and paste continuously.

For a homotopy, use the cover of I^2 by $h^{-1}(U_r)$. Choose a square grid whose cells have diameter less than its Lebesgue number. Extend the given bottom lift across cells, row by row, from left to right. On a cell choose $b_r \circ h + k$ to match the assigned value at its lower-left corner. The only previously assigned points of a new cell are on its bottom and left edges. Along each already assigned bottom or left edge, the two lifts differ by a continuous integer-valued function. That function is constant by connectedness, and zero at the corner. Thus the new piece agrees on shared edges and pastes continuously.

Finally, two lifts on I or I^2 differ by a continuous integer-valued function. Such a function on a connected set is constant: a noninteger threshold would separate two different image values. Agreement at one point therefore proves uniqueness. ∎

The figure shows one extension step of the grid construction.

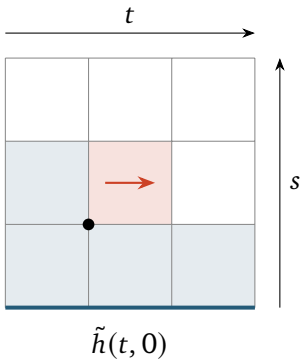


Figure 3.4: Previously assigned cells are shaded. A branch on the next cell is shifted by an integer to agree at the marked corner and hence on the shared edges.

For any loop γ in the circle, define its *winding number* by

$$\operatorname{wind} \gamma = \tilde{\gamma}(1) - \tilde{\gamma}(0) \in \mathbb{Z}.$$

Different initial lifts differ by an integer, so this number is independent of that choice. In a free homotopy of loops, lift the whole square. The endpoint difference is a continuous integer-valued function of s, so it is constant. Thus winding is invariant even when the base point moves.

Theorem 3.7 (Fundamental group of the circle). *Winding gives an isomorphism* $\operatorname{wind}\colon \pi_1(S^1, 1) \to \mathbb{Z}$. *The loop* $t \mapsto p(mt)$ *has winding* m *for every* $m \in \mathbb{Z}$.

Proof. Winding is invariant under free homotopy, hence under endpoint-fixed homotopy. It therefore defines a map on classes. Choose a lift starting at 0. If it ends at m, the homotopy

$$\tilde{h}(t, s) = (1 - s)\tilde{\gamma}(t) + smt$$

fixes the endpoints 0 and m. Its projection gives an endpoint-fixed homotopy from γ to $p(mt)$. Thus winding determines the class, and every integer occurs. For a product of loops with windings m and n, join the first lift to the second lift shifted by m. The joined lift ends at $m+n$, so winding preserves multiplication as addition. ∎

Counter-clockwise travel $p(t)$ has winding +1, fixing the sign convention.

3.4 Retractions and homotopy invariance

Definition 3.8 (Retraction and deformation retraction)**.** A *retraction* onto $A \subseteq X$ is a continuous map $r\colon X \to A$ with $r|_A = \mathrm{id}_A$. A *deformation retraction* also has a homotopy from id_X to $i \circ r$, where $i\colon A \to X$ is inclusion, that keeps every point of A fixed throughout.

Proposition 3.9 (Deformation retractions preserve the group)**.** *For a deformation retraction and $a_0 \in A$, the induced maps r_* and i_* are inverse isomorphisms of fundamental groups.*

Proof. We have $r \circ i = \mathrm{id}_A$. The other composite is homotopic to id_X while fixing a_0. Composing this homotopy with any loop at a_0 is an endpoint-fixed path homotopy. Hence $i_* r_* = \mathrm{id}$ as well as $r_* i_* = \mathrm{id}$. ∎

Identify $\mathbb{C}$ with $\mathbb{R}^2$. For example, on $\mathbb{C} \setminus \{0\}$ the formula

$$h(z,s) = \left(1 - s + \frac{s}{|z|}\right) z$$

deforms the identity to $r(z) = z/|z|$. Its coefficient is positive, so the image never reaches 0, and it fixes every point of S^1. Consequently $\pi_1(\mathbb{R}^2 \setminus \{0\}, 1) \cong \mathbb{Z}$. For a loop in the punctured plane, its winding means the winding of its normalization $\gamma/|\gamma|$.

A nonempty space is *simply connected* if it is path-connected and all based loops are endpoint-fixed homotopic to the constant loop.

A space is *contractible* if its identity is homotopic to a constant map with value x_* in the space.

Proposition 3.10 (Contractible spaces)**.** *Every contractible space is simply connected.*

Proof. The contraction gives paths from all points to x_*, so the space is path-connected. For a loop γ at x_0, it also gives a free homotopy to e_{x_*}. The moving-basepoint construction above converts this to a based homotopy from γ to $a \cdot e_{x_*} \cdot \bar{a}$, which cancels to e_{x_0}. ∎

Maps $f\colon X \to Y$ and $g\colon Y \to X$ give a *homotopy equivalence* if $g \circ f \simeq \mathrm{id}_X$ and $f \circ g \simeq \mathrm{id}_Y$. The general theorem that such f induces an isomorphism on fundamental groups at every base point is cited from Hatcher, Proposition 1.18. The deformation-retraction case used here was proved directly above.

3.5 No retraction of the disc onto its circle

Let $D^2 = \{z \in \mathbb{C} : |z| \le 1\}$ be the closed disc.

Corollary 3.11. *There is no retraction $r\colon D^2 \to S^1$.*

Proof. The homotopy $(z, s) \mapsto (1 - s)z$ contracts the disc. Thus $\pi_1(D^2, 1)$ is the trivial group, consisting only of its identity. If r were a retraction and $i\colon S^1 \to D^2$ the inclusion, then $r \circ i = \mathrm{id}_{S^1}$. Therefore $r_* i_*$ is the identity of $\pi_1(S^1, 1) \cong \mathbb{Z}$. But i_* takes values in the trivial group, so the composite sends every class to the identity element. It cannot be the identity map of this nontrivial group. ∎

The base point here is 1 on the circle. Contractibility gives the trivial group there even though the displayed contraction ends at the centre.

3.6 The fundamental theorem of algebra

Corollary 3.12 (Fundamental theorem of algebra)**.** *Every complex polynomial of degree at least one has a complex root.*

Proof. Write $f(z) = a_n z^n + \sum_{j=0}^{n-1} a_j z^j$ with $n \geq 1$ and $a_n \neq 0$. Choose

$$R > \max\left\{1, \frac{\sum_{j=0}^{n-1} |a_j|}{|a_n|}\right\}.$$

For $|z| = R$ the lower terms satisfy

$$\sum_{j=0}^{n-1} |a_j| R^j \leq R^{n-1} \sum_{j=0}^{n-1} |a_j| < |a_n| R^n.$$

Thus the loops

$$a_n R^n p(nt) + s \sum_{j=0}^{n-1} a_j R^j p(jt)$$

avoid zero for every $s \in I$. Normalizing gives a free homotopy of circle loops from the leading term to $f(Rp(t))$. Choose $b \in \mathbb{R}$ with $p(b) = a_n/|a_n|$. The leading loop has phase $b + nt$, so its winding is n.

The figure illustrates this comparison for a quadratic.

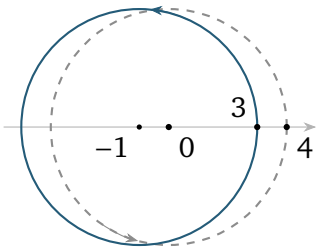


Figure 3.5: For $f(z) = z^2 - 1$ and $R = 2$, the comparison loops are $4p(2t) - s$, each traversed twice. Dashed: $s = 0$, with starting point 4; solid: $s = 1$, centred at -1, with starting point 3.

Now suppose f had no zero anywhere in $\mathbb{C}$. The map $h_f\colon I^2 \to \mathbb{C} \setminus \{0\}$ given by

$$h_f(t, s) = f((1 - s)Rp(t))$$

would be a homotopy of nonzero loops from $f(Rp(t))$ to the constant $f(0)$. These families may move the base point. After normalization, free-homotopy invariance would preserve winding and give its final value zero. This contradicts its value $n \geq 1$. Hence f has a root. ∎

Self-check. Compute $\pi_1(S^1 \times S^1, (1,1))$ using the coordinate projections. Prove the group identification, not just its surjectivity.

Answer. Send $[(\gamma, \delta)]$ to $(\operatorname{wind}\gamma, \operatorname{wind}\delta) \in \mathbb{Z}^2$. Projecting a product homotopy gives endpoint-fixed homotopies of the coordinates. Winding is invariant under these, so the map is well-defined. Projection respects concatenation, so the map preserves multiplication coordinatewise. Conversely, the component test of Lecture 1 makes a pair of loops a loop in the product, and a pair of endpoint-fixed homotopies a homotopy in the product. Every $(m,n) \in \mathbb{Z}^2$ is represented by $t \mapsto (p(mt), p(nt))$. If both windings vanish, each coordinate loop has an endpoint-fixed contraction; their pair contracts the product loop. If two classes have the same image, their product with the inverse of the second has both windings zero, so is trivial. The two classes therefore agree. The group is $\mathbb{Z}^2$.

Summary. Classes of loops at one fixed base point form the fundamental group. Changing the base point along a path gives an isomorphism that can depend on that path. In a path-connected space, free homotopy keeps only conjugacy classes. Lifting computes $\pi_1(S^1, 1) \cong \mathbb{Z}$; deformation retraction gives the same group for the punctured plane. This integer rules out a disc-to-circle retraction and proves the fundamental theorem of algebra. Lecture 4 extends the lifting construction to other maps.

Reading. Hatcher, *Algebraic Topology*, §1.1, pp. 25–38 (Theorem 1.8 and Proposition 1.18); Munkres, *Topology*, §§51–56, §58 and Lemma 27.5.